

Jihye (Jessica) Kim

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SUMMARY

LLM safety and alignment researcher with a Ph.D. from KAIST and 3 years at Deloitte Consulting. First-author publication in *Information Systems Research* (UTD24/FT50); papers accepted at TrustNLP and SemEval workshops at ACL 2026, and submitted to the AIWILD Workshop at ICML 2026. Applies econometric identification methods—staggered DID, IV, Double ML, Causal Forest—to LLM evaluation and agent safety: moving beyond benchmark accuracy to measure *why* and *for whom* safety interventions work or fail.

RESEARCH INTERESTS

LLM Safety & Alignment · Agent Deception & Robustness · Causal Inference for AI Evaluation · Large-scale Observational & Experimental Methods · Societal and Economic Impacts of Information Systems

SKILLS

Causal Inference & Econometrics: Causal ML, Staggered DID, PSM/IPTW, IV, RD, SCM, A/B Testing, Experimental Design

NLP & LLMs: LLM Fine-tuning (LoRA/PEFT), RAG, Prompt Engineering, Agent Architecture, LLM Evaluation & Safety

ML & Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost

Programming & Data: Python (Pandas, NumPy), R, SQL, Git

EDUCATION

University of California, Santa Cruz

M.S. in Natural Language Processing · Dept. of Computer Science and Engineering · Santa Clara, CA

Expected Jan 2027

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. in Information Systems · Dept. of Management Engineering · GPA 4.1/4.3 · Seoul, Korea

2021–2025

Sogang University

B.S. in Mathematics — Highest Distinction · Full Tuition Scholarship · Seoul, Korea

RESEARCH EXPERIENCE

UC Santa Cruz — Graduate Researcher

NLP & LLM Safety · Santa Clara, CA

2025–Present

- **Deceptive Bypass in Safety-Filtered LLM Agents** — Problem: output-level safety filters are widely assumed to curtail harmful agent behavior, but the underlying drive to maximize utility may simply reroute through unmonitored channels. Used a randomized intervention framework on the CaSiNo benchmark (N=500) with causal identification to isolate filter effects. Found that while the Appraisal Gate reduced overt coercion from 35% to 6%, preference hallucination returned to vanilla-baseline levels (33–37%). Key insight: K-Level reasoning incidentally suppresses hallucination; gating coercion releases this suppression. Implication: surface-level filtering is insufficient—alignment must target the agent's objective function. Accepted at **TrustNLP @ ACL 2026** (single author). [\[GitHub\]](#) [\[Paper\]](#)
- **MIRAGE: Pre-Output Detection of Covert Agent Exfiltration** — Problem: LLM agents coerced into encoding sensitive data (Base64, ROT13, acrostic, etc.) evade all output-side detectors. Investigated whether the underlying computation leaves a detectable trace in model internals. Identified a shared low-dimensional subspace in the residual stream that activates across nine encoding families; probe trained on eight families generalizes to the held-out ninth at **AUC 0.975–1.000**. Critical insight: the subspace flips polarity at the planning token—distinguishing inline execution from tool-delegation *before* encoded text exists, enabling pre-output interception. Real-time two-channel monitor achieves **AUC = 0.918** vs. 0.518 for output-only detection. Submitted to **ICML 2026 Workshop AIWILD** (Submission #244).
- **Relational Displacement Benchmark** (in progress) — Constructing a benchmark to evaluate whether LLMs maintain consistent identity and role-awareness across multi-turn contexts where implicit relationships shift—a stress test for pragmatic reasoning that current safety evals do not cover.
- **Epistemic Honesty & Verbal-Probabilistic Alignment** (in progress) — Problem: LLMs routinely verbalize high confidence while internal distributions signal uncertainty—a calibration failure with direct safety implications in

high-stakes deployment. Applying causal intervention design to evaluate which structural prompt changes force alignment between logit-level uncertainty and expressed confidence. [\[GitHub\]](#)

- **LLM Robustness to Social Pressure & Adversarial Persuasion** (in progress) — Problem: agents in agentic pipelines are susceptible to social nudges and authority cues that override alignment-critical decisions without triggering safety filters. Designing controlled experiments to identify which persuasion mechanisms most reliably compromise agent independence, and testing targeted mitigations.
- **Multi-Turn RAG for Conversational AI** — Built hybrid dense-sparse retrieval pipeline (BGE-M3 + BM25) with conversational query rewriting; achieved **+44.3% Recall@10** over zero-shot baselines across 4 QA domains. Accepted at **SemEval 2026 @ ACL 2026**.

KAIST — Graduate Researcher

2021–2025

Information Systems & Causal Inference · Advisor: Wonseok Oh, Ph.D. · Seoul, Korea

- **On-Demand Wage Access & Financial Well-Being** — Estimated causal effects of earned wage access on **4,000 low-wage workers** using staggered DID, PSM, IV, and Double ML on 18-month longitudinal transaction data. Found **+12.9% increase in financial monitoring duration** and **+3.7% increase in saving frequency**; validated via randomized online experiment (Prolific, n=312). Published in **Information Systems Research** (ISR, UTD24/FT50), 2026. [\[Journal\]](#) [\[Media\]](#)
- **Self-Order Kiosk Adoption & Demand Variety** — Measured causal impact of kiosk deployment using staggered DID and IPTW on 17-month store-level panel data (**2,000+ stores**). Found **+13.6% increase in customization behavior** with asymmetric effects by regional income. Under revision; accepted at **WISE 2024**. [\[Slides\]](#)
- **Google In-App Payment Mandate & Platform Governance** — Analyzed anticipatory behavioral responses using app-week DID, triple-differences, and PSM on **35,000-user panel data**. Identified demand reallocation by income tier and app rank. Under review at **JMIS** (FT50).
- **Trending Keyword Removal & Search Behavior** — Applied DID and structural MDCEV demand model to Nielsen Mobile App Panel data to disentangle substitution patterns, attention effects, and information inequality. Accepted at **HICSS 2024**. [\[Paper\]](#)
- Teaching: Lecturer, Business Analytics (32 undergraduates, eval 92.67/100, Kyung Hee University); Teaching Assistant for 7 MBA/EMBA courses at KAIST (Database, Financial Big Data, Cloud Computing, and others).

INDUSTRY EXPERIENCE

Deloitte Consulting LLC — Senior Consultant

Jan 2018–Feb 2021

Management Consultant (Full-Time Employee) · Seoul, Korea

- Designed KPI frameworks and performance management systems for C-suite executives across semiconductor and chemical industries; built Oracle Cloud-integrated analytics dashboards to track real-time KPI attainment.
- Built statistical models on **5,000+ employee records** to diagnose organizational culture, workforce behavior, and productivity drivers for large-scale digital transformation projects.
- Led HR due diligence for a global e-commerce company's cross-border M&A; synthesized workforce analytics and organizational risk assessments for executive decision-making.

PUBLICATIONS & REFEREED CONFERENCES/WORKSHOPS (* = first author)

NLP & LLM Research

- Kim* (Single Author). "Coercion Suppression & Preference Hallucinations in LLM Negotiation Agents." *TrustNLP @ ACL 2026* — **Accepted**. [\[GitHub\]](#) [\[Paper\]](#)
- Revankar, Chauhan, Kim et al. "MIRAGE: A Polarity-Flipping Encoding Subspace in LLM Agents." *ICML 2026 Workshop AIWILD* — **Submitted** (Submission #244).
- Revanka, Kim et al. "Multi-Turn RAG Retrieval for Conversational AI." *SemEval 2026 @ ACL 2026* — **Accepted**.
- Kim* et al. "Bridging the Confidence Gap: Structural Prompting for Verbal-Probabilistic Alignment." *In Preparation*. [\[GitHub\]](#)
- Kim et al. "Relational Displacement Benchmark for LLMs." *In Progress*.

Information Systems & Causal Inference

- Kim* et al. "Working Daily, Paid Monthly? Effects of On-Demand Earned Wage Access on the Financial Well-Being of Low-Wage Workers." *Information Systems Research (ISR)*, UTD24/FT50, 2026 — **Published**. [\[Journal\]](#) [\[Media\]](#)
- Kim* et al. "The Economics of In-App Payment Options: Implications for Digital Platform Governance." *Journal of Management Information Systems (JMIS)*, FT50 — **Under Review**.
- Kim* et al. "Beyond Efficiency: The Impact of Self-Order Kiosk Adoption on Demand Variety." *Manufacturing & Service Operations Management (MSOM)*, UTD24/FT50 — **In Revision**.
- Kim*, Yoon. "Could Self-Order Kiosks Drive Unequal Sales Variety?" *WISE 2024*, Bangkok, Dec. 2024 — **Accepted**. [\[Slides\]](#)
- Park, Kim et al. "The Information Billboard: Effects of Popular Search Terms on Search Behaviors and Digital Divide." *HICSS 2024*, Manoa, HI, Jan. 2024 — **Accepted**. [\[Paper\]](#)
- Kim* et al. "Working Daily, Paid Monthly?" *CIST 2023*, Phoenix, AZ — **Accepted**.
- Kim* et al. "The Economics of In-App Payment Options." *CIST 2022*, Indianapolis, IN — **Accepted**. [\[Slides\]](#)
- Kim* et al. "In-App Payment Regulation and Platform Governance." *KrAIS Summer Workshop 2023*, Seoul — **Accepted**.
- Kim* et al. "On-Demand Earned Wage Access and Financial Well-Being." *MSI DEI 2023*, Dallas, TX — **Accepted**.

HONORS & AWARDS

- Ph.D. Fellowship · Korea Advanced Institute of Science and Technology (KAIST), 2021–2024
- Travel Grant Award · KAIST, 2023
- Full Tuition Scholarship for Admission with Highest Distinction · Sogang University
- Teaching Scholarship · Samsung Dreamclass, 2015

ACADEMIC SERVICE

- **Reviewer:** International Conference on Information Systems (ICIS) 2022, 2023, 2024
- **Reviewer:** Conference on Information Systems and Technology (CIST) 2023, 2025
- **Reviewer:** Pacific Asia Conference on Information Systems (PACIS) 2023